UNIT - I :- Activity Questions :-

Q.1. Find the mean, median and mode of : 5, 7, 7, 9, 11.

(i) Mean.

$$\text{Mean} = \frac{5+7+7+9+11}{5} = \frac{39}{5} = 7.8$$

Middle Value → 7 (Median).

Mode → 7 (Frequently occurring).

Q.2 Find the mean, median, mode and mid-Range of: 12, 15, 15, 18, 24.

1. Mean:-

$$\text{Mean} = \frac{\text{Sum of all Values}}{\text{Number of values}}$$

$$= \frac{12+15+15+18+24}{5}$$

$$= \frac{84}{5} = 16.8$$

$\therefore$ Mean = 16.8

2. Median:-

The data is already arranged:- 12, 15, 15, 18, 24

So the median value is 15.

$\therefore$ Median = 15

3. Mode:-

The value occuring most frequently is 15.

15 occurs twice.

$\therefore$ Mode = 15

4. Mid Range:-

$$\text{Mid-Range} = \frac{\text{Maximum} + \text{Minimum}}{2}$$

$$= \frac{24+12}{2} = \frac{36}{2} = 18$$

Q.3. Find the mean, variance and standard deviation of : 4, 6, 8, 10, 12.

1. Mean :- $(\bar{x})$

$$\bar{x} = \frac{4+6+8+10+12}{5}$$

$$= \frac{40}{5}$$

$$\boxed{\bar{x} = 8}$$

2. Variance :-

x	$x-8$	$(x-8)^2$
4	-4	16
6	-2	4
8	0	0
10	2	4
12	4	16
	Total ⇒	40

Variance :-

$$\sigma^2 = \frac{\Sigma (x-\bar{x})^2}{N}$$

$$= \frac{40}{5} = 8.$$

$$\boxed{\therefore \text{Variance} = 8}$$

3. Standard Deviation :-

$$\text{Std. Deviation} = \sqrt{\text{Variance}}$$

$$= \sqrt{8}$$

$$\text{Std. Deviation} = 2.83.$$

Q.4. Find the mean, median, variance, and Standard Deviation of : 8, 10, 12, 14, 16.

1. Mean :- $(\bar{x})$

$$\bar{x} = \frac{8 + 10 + 12 + 14 + 16}{5}$$

$$= \frac{60}{5}$$

$$\bar{x} = 12.$$

2. Median :-

8, 10, 12, 14, 16

So, Median = 12

3. Variance :-

Mean = 12.

x	$x-12$	$(x-12)^2$
8	-4	16
10	-2	4
12	0	0
14	2	4
16	4	16
	Total ⇒	40

Variance,

$$\sigma^2 = \frac{\Sigma (x-\bar{x})^2}{N}$$

$$= \frac{40}{5}$$

$$= 8$$

4. Standard Deviation :-

$$\sigma = \sqrt{8}$$

$$\sigma \approx 2.83$$

Q.6. Find the mean, median, mode, Variance, and Standard deviation of: 3, 5, 5, 7, 10.

1. Mean:- $(\bar{x})$

$$\bar{x} = \frac{3+5+5+7+10}{5}$$

$$\bar{x} = \frac{30}{5}$$

$$\boxed{\bar{x} = 6}$$

2. Median:-

3, 5, [5], 7, 10

↑

Median.

So, $\boxed{\text{Median} = 5}$.

3. Mode:-

5 occurs twice, while the other values occur only once.

So, $\boxed{\text{Mode} = 5}$.

4. Variance :-

mean = 6.

x	$x-6$	$(x-6)^2$
3	-3	9
5	-1	1
5	-1	1
7	1	1
10	4	16
	Total	= 28

Variance,

$$\sigma^2 = \frac{\Sigma (x - \bar{x})^2}{N}$$

5. Standard Deviation :-

$$\sigma = \sqrt{5.6}$$

$$\sigma \approx 2.37$$

Q.6. Find the mean, median and mid-range of: 18, 20, 22, 24, 26.

1. Mean :-

$$\bar{x} = \frac{18 + 20 + 22 + 24 + 26}{5}$$

$$\bar{x} = \frac{110}{5}$$

$$\boxed{\bar{x} = 22}$$

2. Median :-

18, 20, [22], 24, 26

↑

Median = 22

3. Mid-Range :-

Formula:

$$\text{Mid-Range} = \frac{\text{Minimum} + \text{Maximum}}{2}$$

$$= \frac{18 + 26}{2}$$

$$= \frac{44}{2}$$

$$\boxed{\text{Mid-Range} = 22}$$

Q.7. Find the mean, mode, Variance, and Standard deviation of: 6, 8, 8, 10, 12.

Gvn:-

6, 8, 8, 10, 12.

1. Mean:-

$$\bar{x} = \frac{6 + 8 + 8 + 10 + 12}{5}$$

$$= \frac{44}{5}$$

$$\boxed{\bar{x} = 8.8}$$

2. Mode:-

The value 8 occurs twice, while the others occur once.

$$\boxed{\text{Mode} = 8.}$$

3. Variance:-

Mean = 8.8.

x	$x - 8.8$	$(x - 8.8)^2$
6	-2.8	7.84
8	-0.8	0.64
8	-0.8	0.64
10	1.2	1.44
12	3.2	10.24
		Total: 20.80

Variance,

$$\sigma^2 = \frac{20.80}{5}$$

$$\boxed{\sigma^2 = 4.16}$$

4. Standard Deviation :-

$$\sigma = \sqrt{4.16}$$

$$\boxed{\sigma \approx 2.04.}$$

Q 8. Find the mean, median, and variance of: 15, 18, 21, 24, 27.

Soln :- 15, 18, 21, 24, 27.

1. Mean ($\bar{x}$) :-

$$\bar{x} = \frac{15+18+21+24+27}{5}$$

$$= \frac{105}{5}$$

$$\boxed{\bar{x} = 21}.$$

2. Mode :-

2. Median :-

15, 18, **21**, 24, 27

So, Median = 21.

3. Variance:-

Mean = 21

x	$x-21$	$(x-21)^2$
15	-6	36
18	-3	9
21	0	0
24	3	9
27	6	36

Total : 90

Population Variance,

$$\sigma^2 = \frac{90}{5}$$

$$\boxed{\sigma^2 = 18}$$

Q.9. Find the mean, median, and Standard deviation of : 9, 11, 11, 13, 15.

Gvn:- 9, 11, 11, 13, 15.

1. Mean:

$$\bar{x} = \frac{9+11+11+13+15}{5}$$

$$= \frac{59}{5} = 11.8$$

2. Median:-

9, 11, 11, 13, 15

Median

So, Median = 11.

3. Mode:-

11 occur twice

So, Mode = 11.

4. Std. Deviation:-

Mean = 11.8.

x	$x - 11.8$	$(x-11.8)^2$
9	-2.8	7.84
11	-0.8	0.64
11	-0.8	0.64
13	1.2	1.44
15	3.2	10.24

Total = 20.80.

Variance,

$$\sigma^2 = \frac{20.80}{5} = 4.16.$$

$\therefore \quad \sigma = \sqrt{4.16}$

$$\boxed{\sigma \approx 2.04}$$

Q.10. Find the mean, median, mode, mid-Range, Variance, and Std. deviation of: 10, 12, 14, 14, 200.

Given:- 10, 12, 14, 14, 200.

1. Mean ($\bar{x}$):-

$$\bar{x} = \frac{10 + 12 + 14 + 14 + 200}{5}$$

$$= \frac{250}{5}$$

$$\boxed{\bar{x} = 50}$$

2. Median:-

10, 12, 14, 14, 200

So, median = 14.

3. Mode:-

14 occurs twice.

$$\boxed{\text{Mode} = 14}$$

4. Mid-Range:-

$$\text{Mid-Range} = \frac{\text{Min.} + \text{Max.}}{2}$$

$= \frac{10 + 200}{2} = \frac{210}{2} = 105$ //

5. Variance :-

Mean = 50

x	$x - 50$	$(x-50)^2$
10	-40	1600
12	-38	1444
14	-36	1296
14	-36	1296
200	150	22500

Total = 28136.

Variance,

$$\sigma^2 = \frac{28136}{5}$$

$$\boxed{\sigma^2 = 5627.2}$$

6. Std. Dev^n :-

$$\sigma = \sqrt{5627.2}$$

$$\boxed{\sigma = 75.01}$$